

JEFF ANGINA MBOYA

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EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Bachelor of Science in Mechatronics Engineering; Second Class Upper Honours

Nairobi, Kenya

Sept 2017 – Jun 2023

- **Thesis:** *Autonomous Rover for Adaptive Bird Deterrence*

WORK EXPERIENCE

Abstract Machines, France | AI Infrastructure, Secure Compute, Edge Systems

Feb 2026 – Present

Software Engineer II

Remote

- Unlocked a new enterprise market segment by validating AI inference within secure enclaves, achieving a minimal **10%** performance overhead to provide the "verifiable security" required by Tier-1 clients.
- Expanded the Total Addressable Market (TAM) for edge AI by engineering Propeller, a lightweight orchestration layer that achieved **95%** compatibility across low-resource ARM devices.

Octavia Carbon, Kenya | Climate Tech, Cost Optimization, Systems Engineering

Jun 2023 – Feb 2026

Deputy Head of Research & Development Team

On Site, Nairobi, Kenya

- Co-developed technical R&D roadmap for the DAC+Storage plant providing due diligence framework that secured **\$3.1M** in series A funding.
- Generated a **4.6x** return on a **\$137K R&D** budget by hitting key R&D milestones, catalyzing **\$630K+** in follow-on capital to deploy a 9-unit regional pilot fleet in Rift Valley.
- De-risked Global South's first commercial DAC deployment by directing a **5-person** team to **0% downtime** across **400+ test cycles** on a **2-ton/year** pilot — directly validating the technology case for a **30x scale-up** to a **60-ton/year** field unit.
- Diagnosed regeneration cycle time as the binding constraint on unit economics; led **3 cross-functional teams** to implement a redesign that cut cycle time **83%** and increased daily CO₂ throughput **3.6x**.
- Eliminated commercialization barrier by leading a 7-person cross-functional team to execute a build-vs-buy analysis that replaced third-party controllers with a proprietary stack, reducing CAPEX by **63%** and improving long-term unit economics.
- Resolved a **2-year** strategic technical bottleneck by achieving a **99.9%** product purity standard, directly enabling the first-ever commercial-scale deployment of the technology.
- Developed a Gen-3 technology roadmap identifying lightweight composites and microwave-based regeneration as the critical path to a **41%** BOM cost reduction next commercialization phase.

HONORS & AWARDS

- **PCT Patent Grants | World Intellectual Property Organization (2025):** Secured **3 international PCT patents** protecting core technologies critical to achieving sub-\$100/ton direct air capture at commercial scale.
- **Regional Winner | IBM Call for Code (Middle East & Africa, 2021):** Won the MEA region out of **500,000+ developers across 180 nations**, securing **\$5,000** and IBM technical partnership support.

LEADERSHIP & INVOLVEMENT

Society of Engineering Students (S.E.S.)

May 2020 – Dec 2021

Vice-President

JKUAT, Kenya

- Grew club funding **42x** (Ksh 5k → 210k) through strategic sponsorships with **Microsoft**, Moringa School, and the NCA; leveraged partnerships to source keynote speakers from **Goldman Sachs London**, **Deloitte**, **EY**, and **London Business School** for a career summit attended by **446 students**.
- Scaled mentorship outreach **75%** across **7 schools**, reaching **308 students** and securing **\$70,000 in scholarships** for **35 high-achieving mentees**; pioneered an annual tech summit now in its **5th consecutive year**.

Nakuja Project

Feb 2021 – Nov 2022

Strategic Growth & Publicity Manager

JKUAT, Kenya

- Scaled project membership **31x** (from 6 to 188) and established a 12-week internship program that has since onboarded 300+ students across Kenya.
- Formalized the project's primary partnership with the **Kenya Space Agency (KSA)** and expanded cross-institutional R&D to the University of Nairobi and Kenyatta University.

PUBLICATIONS & RESEARCH

- **First Author** — "Improving Small Pest Bird Detection in YOLOv5s for Autonomous Bird Deterrent Systems" 2022
- **Co-Author** — "Simulation and Airflow Experimentation of a Multi-Layer Adsorbent Chamber for Enhanced Direct Air Capture Efficiency" 2025
- **Co-Author** — "Development of a Solid Propellant Motor for High-Powered Model Rockets" 2024
- **Co-Author** — "Development of Solid Propellant Motor for Low Altitude Model Rockets" 2022